

B.Tech. Degree I Semester Regular Examination November 2023

CE/ME/SE 23-200-0101A & EE/EC/CS/IT 23-200-0101B CALCULUS

(2023 Scheme)

Time: 3 Hours

Maximum Marks: 50

Course Outcomes

On successful completion of the course, the students will be able to:

CO1: Solve ordinary differential equations and linear differential equations of higher orders with constant coefficient and apply them in engineering problems.

CO2: Determine the maxima and minima of multi variable functions.

CO3: Evaluate area as double integrals and volume as triple integrals.

CO4: Illustrate the application and physical meaning of gradient, divergence and curl.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A

(Answer ALL questions)

	(5 × 2 = 10)	Marks	BL	CO	PO
I. (a) Check whether $(1-x)dx - (1+y)dy$ is exact or not.		2	L1	1	1
(b) If $x^y = y^x$, find $\frac{dy}{dx}$.		2	L1	2	1
(c) Evaluate $\int_0^2 \int_1^2 (x^2 + y^2) dx dy$.		2	L2	3	1
(d) Compute the divergence of $\vec{F} = xyz\hat{i} + xyz\hat{j} + xyz\hat{k}$ at $(1,1,1)$.		2	L2	4	1
(e) Show that $\vec{F} = (x+3y)\hat{i} + (y-2z)\hat{j} + (x-2z)\hat{k}$ is solenoidal.		2	L1	4	1

PART B

(4 × 10 = 40)

II. (a) Solve $(D^2 - 2D + 1)y = xe^x \sin x$.	5	L3	1	1
(b) Solve the system $\frac{dx}{dt} = 2y, \frac{dy}{dt} = 2z, \frac{dz}{dt} = 2x$.	5	L6	1	1

OR

III. (a) Solve $(D^2 + 4D + 4)y = \cos 5x + e^{3x}$.	5	L5	1	1
(b) Solve $(x^2 D^2 - 2xD + 2)y = x^3$.	5	L4	1	1

IV. (a) Verify Euler's Theorem for the function $u = ax^2 + 2hxy + by^2$.	5	L3	2	1
(b) If $x = r \cos t$ and $y = r \sin t$. Find $J = \frac{\partial(x,y)}{\partial(r,t)}$. Show that $JJ^t = 1$.	5	L5	2	1

OR

V. (a) Find the maximum and minimum values of $f(x,y) = x^3 y^2 (1-x-y)$.	5	L4	2	1
(b) Find the polar coordinates of the following points: (i) $(\sqrt{2}, \sqrt{2})$ (ii) $(-1, \sqrt{3})$ (iii) $(-\sqrt{3}, -1)$ (iv) $(1, -1)$ (v) $(3, 4)$	5	L4	2	1

(P.T.O)

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		Marks	BL	CO	PO
✓ VI.	(a) Change the order and hence evaluate $\int_0^1 \int_{x^2}^x (2x^2 + y) dy dx$.	5	L3	3	1
	(b) Evaluate the volume of the solid generated by revolving the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ about the X axis.	5	L6	3	1
OR					
VII.	(a) Evaluate the arc length of one arc of the cycloid $x = a(\theta - \sin \theta)$, $y = a(1 - \cos \theta)$.	5	L5	3	1
	(b) Find the volume of the sphere $x^2 + y^2 + z^2 = a^2$ using triple integration.	5	L6	3	1
OR					
VIII.	(a) Show that $\vec{F} = (6xy + z^3)\hat{i} + (3x^2 - z)\hat{j} + (3xz^2 - y)\hat{k}$ is irrotational and find the potential function of $\vec{F}$.	5	L3	4	1
	(b) If $r = \vec{r} $, where $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$. Prove that $\nabla r^n = nr^{n-2}\vec{r}$.	5	L5	4	1
OR					
IX.	(a) Evaluate $\int_C \vec{F} \cdot d\vec{r}$ if $\vec{F} = 3xy\hat{i} - y^2\hat{j}$ where C is the curve in the XY plane given by $y = x^3$ from $(0,0)$ to $(1,1)$.	5	L4	4	1
	(b) Evaluate using Stoke's theorem $\oint_C \vec{F} \cdot d\vec{r}$ where C is a curve in XY plane around the lines $x=0, y=0, x=5, y=5$ and $\vec{F} = x^2\hat{i} + xy\hat{j}$.	5	L4	4	1

Bloom's Taxonomy Levels

L1 = 14.3%, L2 = 9.5%, L3 = 19%, L4 = 23.9%, L5 = 19%, L6 = 14.3%.
